

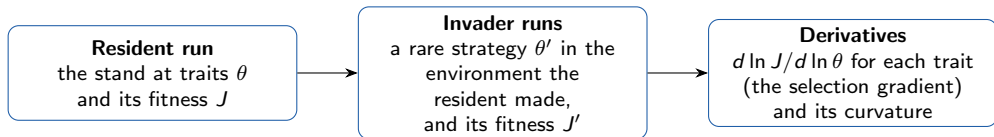
Fitness gradients for TF24 in plant

What changed in plant, odelia and phylloptim, what it costs and what it buys

8 October 2026

An evolutionary analysis needs fitness and its derivatives, many times over

regnans finds and classifies singular strategies from plant runs of three kinds.



- One singular strategy in one trait takes about **84 such runs**. A resident run on a 40-year record takes about 90 s.
- Until now plant returned J alone. Each derivative was a central difference, two runs per parameter, so a TF24 gradient over its 49 parameters cost about **98 runs**.
- An invader ran with its own time steps, so J' moved in small jumps as θ' moved, and differences of J' inherited the jumps.

The whole analysis step, end to end, in plant's R interface

```
p    <- add_strategies(scm_base_parameters("TF24"), trait_matrix(0.32, "lma"))
p$node_schedule_times <- list(seq(0, 39.6, length.out = 108)) # 108 introductions over 40 years

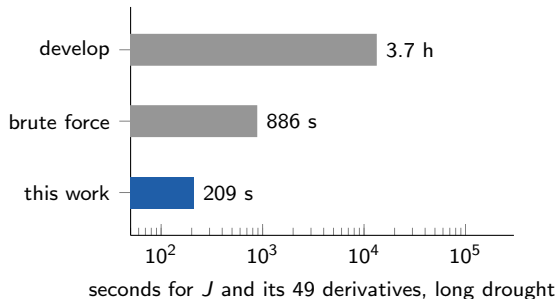
base <- control_tf24(3e-5, Control(node_density_in_birth_date = TRUE,
                                   ode_split_sign_changes = TRUE)) # slides 8, 9, 10
pilot <- run_scm(p54, env, control_tf24(1e-3), record_trajectory = TRUE) # p at 54 introductions
ctrl  <- control_window(pilot, list(lma_half, lma_double), base = base) # slide 11

scm  <- run_scm(p, env, ctrl, record_trajectory = TRUE)
g    <- stand_gradient(scm, "offspring_production") # all 49 derivatives in one pass, slide 6
scm$run_mutant(invader) # an invader on the resident's steps, slide 7
d    <- diagnose_scm(p, env, ctrl, invaders) # error estimate and failures, slide 13
```

On the 40-year long-drought record the middle block takes **209 s**: 37 s for the pilot, 40 s for the resident run and 132 s for its gradient. It returns J and all 49 elasticities. The same answers by brute force take 886 s. Every function above is on the branches; each later slide explains one line.

Timed alone, one run each: docs/measurements/headline/.

The same answers in a quarter to a half of brute force's time



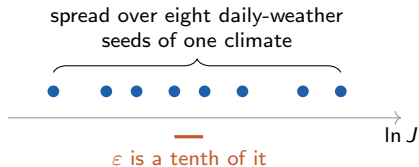
develop plant's develop with its default Control. One run takes 134 s; its gradient needs 100 runs by differences, so this bar is computed.

brute force this work's code with every new control off: a tight tolerance ($1e-5$) and twice the introductions (215). It uses the new gradient pass too.

this work the code on the previous slide, pilot included.

- On the episodic record the ratio is also 0.24 (128 s against 536 s), and all 50 quantities lie within 0.40ε of brute force's.
- On long drought 108 introductions are too few: some elasticities miss by up to 2.1ε . At 215 introductions this work matches brute force's accuracy in 378 s, 0.43 of its time. Slide 14 has the table.

Accuracy is judged against what the weather alone does to the answer

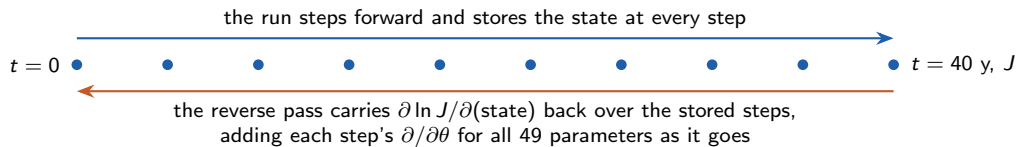


Eight records of the same long-drought climate differ only in the day-to-day weather. Each quantity's spread across them sets its tolerance: ϵ is a tenth of that spread.

quantity	resident	invader
$\ln J$	0.025	0.025
elasticity in <code>lma</code>	0.087	0.20
elasticity in <code>a_dG2</code>	0.019	0.050
any other elasticity	at least 0.01	

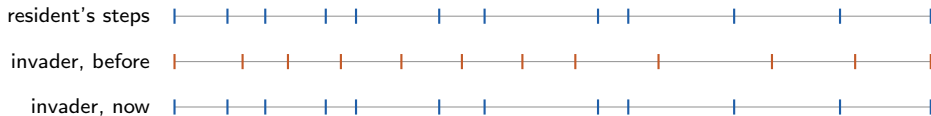
- 0.025 in $\ln J$ is 2.5% in J . An answer within ϵ is off by less than a tenth of what a different year's weather would change.
- A run is *reproducible* when a 5% change of tolerance, or moving the introductions by a quarter of their spacing, moves each quantity by less than $\epsilon/3$.
- Every claim is checked on five rainfall records: constant rain, long-wet, episodic, dry and long drought.

One reverse pass gives every parameter's derivative for 2.5 runs' cost



- `stand_gradient(scm)` runs that pass over the steps that `run_scm(..., record_trajectory = TRUE)` stored. It costs 2.5–2.7 runs at any number of introductions, against about 98 runs by central differences.
- It agrees with central differences to $7\text{e-}4$ of the resident's `lma` elasticity, and closer for the others checked.
- The derivative's arithmetic is recorded and replayed by XAD, vendored in `odelia`. Only doubles cross into R; the derivative types live inside the one call.

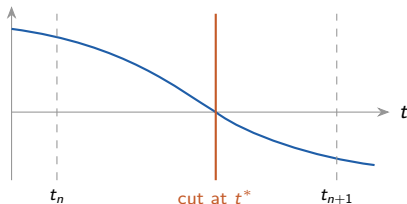
An invader runs on the resident's recorded steps, so J' moves smoothly



- `run_scm` records each accepted step. `scm$run_mutant(invader)` then integrates the invader on exactly those steps, in the light and water the resident made.
- With its own steps, an invader's J' moved a little each time the step pattern changed. On the recorded steps J' is a smooth function of θ' , so its differences and its own reverse pass are meaningful.
- At the resident's traits $J' = J$ to the last bit, which the tests check on every record.
- The first invader reuses the resident's recorded run instead of repeating it: 139 s against 300 s.

Where a cohort's net production changes sign inside a step, its step is cut there

a cohort's net production



TF24 grows a cohort by the positive part of its net production, so the cohort's rate has a corner where net production crosses zero. A Runge–Kutta step integrates a smooth rate well and a corner badly. As θ moves, the corner slides past the step's internal stages, and J moves in small jumps: the gradient becomes a staircase.

With `ode_split_sign_changes = TRUE`, plant finds the crossing on the step's own interpolant and integrates that cohort in two pieces, cut there. The other cohorts and the soil keep the whole step.

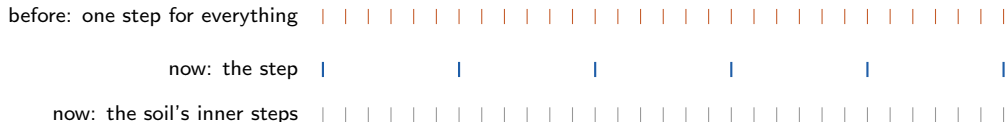
- In J lies $3\text{--}13\times$ nearer the converged answer wherever cohorts cross zero.
- A 5% tolerance change moves no gradient entry by more than $\varepsilon/6$; without the cut it moved entries by up to 0.52ε . The cut costs 5–9% of a gradient run.

`control_tf24()` names TF24's step settings once, each for a measured reason

setting	why
relative tolerance <code>tol</code> , absolute at 10^{-4} <code>tol</code> soil layers' error tolerance $10\times$ looser	Tied this way, J 's error stays within about half of <code>tol</code> . With plant's default absolute tolerance it was about $20\times$ larger at <code>tol = 1e-4</code> . The soil set most steps while carrying little of J 's error. Loosened tenfold, a gradient run takes 21% less time and both roles still pass the accuracy and reproducibility tests at $3e-5$.
each state's loosening bounded at $100\times$	The window (slide 11) loosens by up to $100\times$ more. Unbounded, the soil's reached $1000\times$, an accepted step could push a soil layer to its potential ceiling, and the gradient was refused there.
steps capped at 15 days	An invader's storage pools go unstable on steps longer than 26 days. On the episodic and dry records, the invader at twice the resident's <code>lma</code> failed on 27-day steps. With the cap none fails.
soil stepped alone below a 10% share	The next slide.

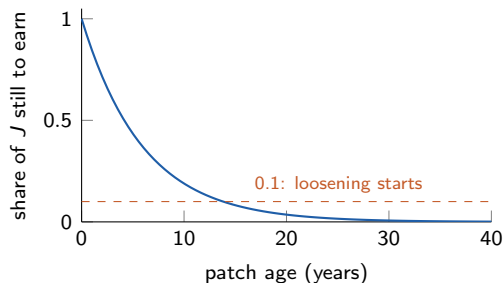
With the window, these settings save 31–34% of a gradient run against the tied tolerance alone.
plant's own `Control()` defaults are unchanged.

Where the plants barely draw on the soil, the soil takes its own short steps



- Water moves through TF24's soil layers within days, and an explicit step must stay short enough for the soil to remain stable. On long drought the soil set 16 447 of 17 684 steps, while the plants could have taken far longer ones.
- Where the plants take under a tenth of the water moving through the soil, the soil hardly depends on them over one step. odelia then integrates the soil alone in short inner steps, under the plants' uptake extrapolated across the step, and the plants set the step itself.
- A gradient run takes 0.63 of the time (376 s against 596 s). Every elasticity moves by less than 0.063ε , and the reverse pass runs through the inner steps.

Once little fitness is left to earn, each step's tolerance is loosened



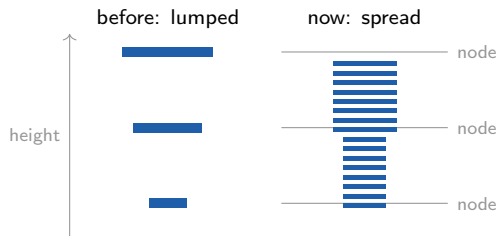
Shape for illustration.

J is earned mostly early in a patch's life. Write $R(t)$ for the share of J still to be earned after t . Once R falls below 0.1, a step starting at t takes its tolerance multiplied by $0.1/R(t)$, up to $100\times$ looser.

`control_window()` reads $R(t)$ off a cheap pilot run, 54 introductions at tolerance $1e-3$, together with the invaders at half and twice the resident's $1ma$. The pilot costs 0.4–0.8 of the run it serves.

- It saves 21–23% of a run's steps on the pulsed records, and 2% under constant rain.
- On all five records nothing fails, and $\ln J$ lies within $5.8e-5$ of a reference at tolerance $1e-6$.

With leaf area spread between nodes, doubling the nodes cuts the error by four



On the birth-date coordinate plant put each node's whole leaf area at that node's height, so the light field stepped at every node. Its error fell with more nodes, but at no fixed rate.

Each interval between neighbouring nodes now spreads its leaf area over eight point crowns between the two nodes' heights. The error then falls with the square of the spacing: halving the spacing divides it by four.

- That makes the error measurable. With Q a quantity from a run and $Q_{1/2}$ the same quantity at half the introductions, the converged answer is near $Q + (Q - Q_{1/2})/3$.
- Measured ratios are 3.5–4.0. The half-introduction run reports 0.87–0.99 of the true error, against 0.33–0.41 before. The spread costs 6.7% of a gradient run.

diagnose_scm() reports each run's error estimate, distance and failures

It repeats the run at every other and at every fourth introduction, with each invader on each, and returns three tables. These rows come from the eight-year test stand at 9, 5 and 3 introductions.

run	quantity	value	half	quarter	error	ratio
resident	ln J	4.7033	4.7979	5.1101	-0.0315	3.30
resident	lma	-1.6028	-1.5512	-1.4264	-0.0172	2.42
late	ln J	5.5647	5.6787	6.0344	-0.0380	3.12

run	stage	message
broken	walk	L_tip must be...

- **error** is $(Q - Q_{1/2})/3$, the amount to add to reach the converged answer. **ratio** is $(Q_{1/2} - Q_{1/4})/(Q - Q_{1/2})$, near 4 where the estimate can be trusted. Nine introductions are too few here, and the ratio shows it.
- **distance** lists each invader's $\ln(\theta'/\theta)$ for each parameter it moves; the invader late is 0.405 away in hmat.
- A run, invader or gradient that throws becomes a row of **failures**, and the analysis continues. The two extra runs cost about three quarters of a run, so it runs once per analysis.

On long drought, 215 introductions match brute force's accuracy at 0.43 of its time

The reference is extrapolated from this work's runs at 108 and 215 introductions, $Q_{215} + (Q_{215} - Q_{108})/3$. Each row compares $\ln J$ and the 49 elasticities with it, in units of each quantity's ε . Time is relative to brute force's.

long drought	$\ln J$	elasticities, median	largest	time
brute force: 215 introductions, 1e-5	0.05ε	0.12ε	0.49ε	1
this work, 215 introductions	0.05ε	0.13ε	0.53ε	0.43
this work, 108 introductions	0.20ε	0.50ε	2.1ε	0.24

- At 108 introductions the remaining error comes from the nodes. The half-introduction run reports it, so a run can tell that it needs more introductions.
- On the episodic record 108 introductions suffice: every quantity lies within 0.40ε of brute force's.
- No run, invader or gradient fails on any of the five records.

What changes for code that uses plant today

- Every new behaviour sits behind a `Control` field or a new function, and is off by default. With it off, results are bit-identical, apart from the two moves below.
- The height coordinate, plant's default, does not move.
- **FF16 on the birth-date coordinate moves**, because the spread changes how leaf area enters the light field. The one-species anchor goes from 17.14 to 18.01, and the two-species test from 12.05 and 16.59 to 12.07 and 16.76. At the default schedule FF16's error is now $9\text{--}25\times$ larger. It falls at the square law, so a half-introduction run estimates it. Each move was registered before the references were blessed again.
- TF24's model version goes to v13. Its numbers change with the gradient pass, a storage pool offset, and a newborn's height found to roundoff.
- Under plant's default `Control()`, TF24's J on long drought is 0.45 on `develop`, against 12.75 converged. On this work's code the same defaults give 0.18, so the default settings, more than the model, carry that gap.

Known gaps

- **Episodic rain needs finer introductions** for the error estimate to hold. Its error falls at the square law only once the spacing is below its dry spells (median 31 days), past 429 introductions.
- **Under constant rain the gradients move by up to 3ϵ** with where the later introductions sit; only the first year is graded finer.
- **A gradient blow-up with the soil stepped with the plants at a loose tolerance.** At tolerance $1e-3$, with the soil's tolerance ten times looser and no inner steps, the resident's elasticities come out near 10^{50} while J is right, and nothing refuses. Stepping the soil alone, now the preset's default, removes it at every tolerance tried. Not yet root-caused.
- **How far one resident's recorded steps can serve** a moved resident or a distant invader is unmeasured; `diagnose_scm()` reports the distance without a bound.
- The branches sit on the fork's `develop` with `odelia 0.5.0`, and need a rebase onto upstream `develop` (`odelia 0.6.2`) before PRs open there.

How to review: four groups, merged in this order

group	branches	read first	its test
1. Gradients	PLANT-93, offspring-adjoint, PLANT-95-99	stand_gradient, run_mutant	the pass against central differences; $J' = J$ to the bit
2. The cut at sign changes	state-weights, PLANT-100-103, odelia#53-54, phylloptim#17	where a step is cut, and how the reverse pass crosses the cut	the 5% tolerance change; the pass against differences
3. Step settings	PLANT-104-106, odelia#55	control_tf24, control_window	replays bit for bit; nothing fails on five records
4. Nodes and diagnostics	PLANT-107-110	the spread in the light field; diagnose_scm	the square law; the FF16 re-bless

Every branch closes one issue on the aornugent forks, and each group is bit-identical with its new controls off. Group 2 is the largest, and changes what odelia hands plant inside a step. Group 4 holds the one decision for maintainers: whether the spread should apply to FF16's birth-date coordinate.